

University of Engineering and Technology Lahore

Department: Computer Science

Section Course Detail	
Semester	Spring 2026
Department	Computer Science
Section	
Subject Title	CMPE-222 Digital Logic Design
Subject Domain	Engineering
Subject Knowledge	Computing
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Measureable Student Learning Outcomes					
CLOs	Description	PLOs	Domain	Domain Level	Assessments
CLO1	Carry out mathematical computations in Boolean algebra and in various number systems (Binary, Octal, Hexadecimal).	PLO01	Cognitive	2. Understand	Quiz1, Assignment
CLO2	Analyze digital circuits in combinational and sequential logic.	PLO02	Cognitive	4. Analyze	Quiz2, Mid term exam
CLO3	Design combinational and sequential logic circuits from functional description of digital systems.	PLO03	Cognitive	3. Apply	Final Exam
CLO4	Comply with plagiarism guidelines	PLO08	Affective	2. Understand	Assignment, Project
Class Timings					

Grading Policy		
Assignments 10		
Mid term exam 30.0		
Quizzes 10		
Final Exam 40.0		
Project 10		
Section Content		
Week (Lec)	Topics	CLO's

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Section Content		
Week (Lec)	Topics	CLO's
week1	Motivation, From Silicon to computers and what part this course covers, Why binary? Binary numbers, Octal and Hexadecimal numbers, Interconversion between different number systems, arithmetic in different number systems, Radix compliments, BCD numbers, arithmetic in BCD, Binary and decimal codes, error detection codes, alphanumeric codes, Binary logic, Logic gates, Noise Margins, Integrated Circuits.	CLO1
week2	Boolean Algebra, Two-valued Boolean algebra and logic gates, Axioms of Boolean algebra (Huntington postulates), Theorems of Boolean algebra, Boolean functions, Truth tables, Logic diagrams, Venn diagrams.	CLO1
week3	Simplification through algebraic manipulation, Compliment of a Boolean function, Canonical and standard forms, positive and negative logic.	CLO2
week4	Simplification through Karnaugh Maps, Two, three and four variable maps, SOP and POS simplification, Handling don't care conditions.	CLO3, CLO2
week5	Prime implicants, Simplification through prime implicants. NAND and NOR implementations, Universal properties of NAND and NOR gates, Multilevel NAND and NOR circuits.	CLO3, CLO2
week6	Properties of the XOR and XNOR functions, Analysis and design of SSI combinational circuits, the half and full adder, Binary adder/subtractor, Code conversion, Generation of parity bits and error checking.	CLO3, CLO2
week7	Analysis and design of MSI combinational logic, BCD to Excess-3 converter, BCD adder, Decimal adder, Multiplexer and Demultiplexer. Decoder and Encoder.	CLO2, CLO3
week9	Programmable Logic Devices such as the PLA (Programmable Logic Array), PAL and the Read Only Memory (ROM).	CLO4, CLO2
week10	Multiplexers for inputs having multiple bits. A small combinational Arithmetic Logic Unit (ALU).	CLO3, CLO2
week11	Sequential Logic, Flip flops and bit storage, Types of flip flops, their electronic circuits, characteristic tables and equations, Triggering of flip flops, Edge and level triggered flip flops, Role of the clock.	CLO2

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Section Course Outline Report

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Section Content		
Week (Lec)	Topics	CLO's
week12	Analysis of synchronous sequential circuits, State tables, State Equations, State Diagrams.	CLO2
week13	Excitation tables and sequential circuit design	CLO3
week14	Design of counters, Binary cells, Design of registers, Random Access Memory (RAM).	CLO3
week15	Design of a small microprocessor and introduction to assembly language.	CLO3